Ling Lok Ngai

Address: Fremont, CA 94536 || Cell: (408) 886-1104 || Email: linglok1230@gmail.com
LinkedIn: www.linkedin.com/in/lockengai || GitHub: https://github.com/loklok104

Summary

Backend Software Engineer with 5+ years of experience designing and maintaining scalable distributed systems using Java, Spring Boot, AWS, and microservices. Strong ownership of complex backend initiatives, delivering reliability and performance improvements while leveraging LLMs to accelerate debugging, documentation, and overall development workflows across teams.

Skills

Languages: Java, Python, SQL, TypeScript
Frameworks: Spring Boot, MyBatis
Cloud & Infra: AWS Lambda, EC2, ECS, S3, API Gateway, CloudFormation, Docker
Databases & Search: MySQL, DynamoDB, Redis, Solr, data analytics
Architecture & Concepts: Microservices, RESTful APIs, Caching, , ETL Pipelines
DevOps & Tools: Git, Gradle, IntelliJ, Unix/Linux, CI/CD, Agile (Scrum), A/B Testing
Soft Skills: problem-solving, mentoring, leadership

Work Experience

Amazon — Remote
Software Development Engineer — Jan 2023 – Present

- Design and maintain backend product search infrastructure using **Java, Spring Boot, Solr, AWS**, and **microservices**, handling millions of user queries daily with high availability and low latency
- Restructured Solr indexing schema (styleId to productId) and integrated nested documents to reduce query load by **10%** and improve facet accuracy.
- Implemented dynamic filtering logic in backend services to auto-facet top brands and product types based on ML-inferred user intent, increasing facet click rate by **1.25%** and reducing explicit re-searches by **0.5%**.
- Simplified shoe width options by mapping technical labels (e.g., 4A, B, SS) to user-friendly categories, increasing discoverability and boosting add-to-cart rate by **1.5%**.
- Create RESTful APIs and event-driven services with fallback and rollback mechanisms improving service resilience and overall platform reliability.
- Own and maintain **CloudFormation/CDK stacks** (Java-based) for search microservices, including ECS/Fargate deployment, ALB, and VPC configurations.
- Collaborate across teams (Catalog, Mobile, QA, Product) to deliver features with end-to-end quality and test coverage (unit test, integration test).

Endliss Technology Inc. — Hayward, CA
Software Engineer — Sep 2020 – Jan 2023

- Implemented and designed applications using full **Software Development Life Cycle** (SDLC) that includes Requirements Analysis, Design, Coding, Testing.
- Developed and managed an order information system with report function using **Java**, **Spring**, **Hibernate, AWS** to ease the users looking up, tracking, and recording the orders which decreased the negative feedback by **10%**.

Projects

Search Platform Optimization - Solr Nested Documents Integration

- Restructured Solr indexing from styleId to productId to support better grouping, improve swatch rendering, and boost query performance.
- Integrated nested document modeling to shift group computation from runtime to indexing time, **reducing query load by 10%**.
- Optimized query logic with parent-child joins and custom relevancy scoring to improve top-result accuracy and facet filtering.
- Redesigned core search logic to reduce latency and increase precision for variant-rich products (e.g., shoes with multiple sizes/colors).

Streamline Shoe Width Values for UX improvement

- Mapped technical shoe width labels (e.g., SS, 4A, B) to customer-friendly categories (Narrow, Medium, Wide).
- Implemented new mapping logic in Solr, ensuring backward compatibility and inventory consistency.
- Coordinated with Catalog, Frontend, Mobile, and SEO teams to ensure URL structures and parameter changes did not negatively impact search engine indexing or rankings.
- A/B tested the rollout and ensured rollback readiness across platforms.
- Resulted in a **2% drop in facet click rate** and **1.5% increase in add-to-cart rate**, indicating improved findability and decision confidence.

Search Result Precision via Auto-Faceting for Brands & Product Type

- Enhanced search relevance by dynamically injecting brand/product-type filters into backend microservice logic for brand-intent queries.
- Partnered with the ML team to curate and update a list of top 100 high-traffic brands.
- Reduced noise in ambiguous keyword matches by applying runtime filters based on brand signal confidence.
- Launched A/B testing via Weblab and observed a **1.25% increase in facet clicks**, signaling higher engagement, and a **0.5% decrease in explicit re-searches**, indicating better first-result relevance.
- Contributed to long-term UX direction toward ML-enhanced, intent-aware search experiences at scale.

Education

Master of Science in Computer Science 2015 - 2016
International Technological University - San Jose, CA

Bachelors of Science, Business Administration 2012 - 2013
California State University, East Bay - Hayward, CA